Project Proposals: Uncertainty in Diffusion and Flow Matching Models

1 Per-Timestep Uncertainty in DiT: Mode Selection vs Within-Mode Variability

Project name: Per-Timestep Uncertainty in DiT: Mode Selection vs Within-Mode Variability
Target model: DiT-based models or DiT-based models trained with TREAD.

Short Description

Diffusion Transformer (DiT) models process generation as a sequence of denoising steps, but different timesteps play fundamentally different roles: early steps (high σ) determine global structure — composition, layout, semantic mode — while late steps (low σ) refine local details within an already-chosen mode. This project studies whether these two regimes correspond to measurably different types of uncertainty, and whether uncertainty signals at specific timesteps predict final generation quality.

Vanilla DiT If we work with standard DiT-based models, rather than applying MC Dropout (which is inappropriate for standard DiT architectures trained without dropout), the project uses two non-parametric, inference-time uncertainty proxies: ensemble variance across multiple noise seeds, and attention entropy within DiT's self-attention layers. No retraining or weight modification is required.

Thread If we work with a DiT trained with **TREAD** (Token Routing for Efficient Architecture-agnostic Diffusion Training) – trained with per-token stochastic layer skipping we can actually work with more standard uncertanty estimation. On such a model, an MC-style epistemic estimate becomes methodologically legitimate, because the network was actually trained as an ensemble of sub-networks rather than having dropout injected post-hoc.

Overall we can work with three complementary, inference-time, training-free uncertainty proxies

1. ensemble variance across noise seeds
2. routing variance across TREAD routes (thread-based DiT)
3. attention entropy

No retraining or weight modification is required; we use publicly available TREAD checkpoints.

Background and Motivation

Why standard DiT resists MC Dropout. Standard DiT-S/B/XL models (e.g., trained on ImageNet-256) use no dropout during training. Injecting dropout into pretrained weights that were never trained with it produces uninterpretable variance estimates, so post-hoc MC Dropout is not a sound epistemic estimator for vanilla DiT.

TREAD as a stochastic-depth training procedure. TREAD (Krause et al., 2025) was introduced as a training-efficiency method: during training, a randomly selected subset of tokens $\mathcal{T}_r \subset \mathcal{T}$ bypasses a contiguous block of layers $L_i, \ldots, L_j$ (replaced by the identity map) via a route $\mathbf{r}_{i\to j}$, and is reintroduced at a deeper layer. The routed denoiser is

$$D_\theta^{\mathbf{r}_{i\to j}} = L_{B-1} \circ \cdots \circ \begin{cases} \mathrm{id}, & \tau_k \in \mathcal{T}_{\mathbf{r}_{i\to j}} \\ L_j \circ \cdots \circ L_i, & \text{otherwise} \end{cases} \circ \cdots \circ L_0,$$

with tokens selected randomly per training sample at a fixed selection rate < 1. TREAD's own analysis states that this introduces additional variance that the network learns the denoising step with a variable number of layers. Functionally, this is per-token stochastic depth: training under a distribution over sub-networks, exactly the regularization regime that makes MC-style epistemic estimation legitimate. TREAD presents this purely as an efficiency mechanism; we are the first to use it as an uncertainty-quantification substrate.

Three inference-time proxies (no weight modification).

- **Ensemble variance** (vary the seed): run the same condition with K initial noise samples $z^{(1)}, \ldots, z^{(K)} \sim \mathcal{N}(0, I)$ through the *full* path, measure variance of $\hat{x}_0$-predictions at each timestep t:
$$U_{\text{ens}}(t, c) = \mathrm{Var}_k \left[D_\theta(x_t^{(k)}, t, c) \right].$$
- **Routing variance (ONLY for THREAD** (vary the sub-network): on a TREAD-trained model, with the seed z *fixed*, draw M random routes $\mathbf{r}^{(1)}, \ldots, \mathbf{r}^{(M)}$ and measure variance of $\hat{x}_0$-predictions:
$$U_{\text{route}}(t, c) = \mathrm{Var}_m \left[D_\theta^{\mathbf{r}^{(m)}}(x_t, t, c) \,\middle|\, z \right].$$
This is variance across sub-networks at fixed data noise — an epistemic signal obtainable without retraining and without an explicit ensemble of weights.
- **Attention entropy** (single pass): Shannon entropy of attention weight distributions in each DiT block at timestep t:
$$U_{\text{attn}}(t, \ell) = -\sum_{i,j} A_{ij}^{(\ell)} \log A_{ij}^{(\ell)},$$
where $A^{(\ell)}$ is the attention map of layer ℓ. High entropy = diffuse attention = model is uncertain which patches are relevant.

Epistemic/aleatoric decomposition. Because routing variance (sub-network noise) and seed variance (data noise) are two *independent* sources, the law of total variance lets us separate them. Define

$$U_{\text{epi}}(t, c) = \mathrm{Var}_{\mathbf{r}}[D_\theta^{\mathbf{r}}(x_t, t, c) \,|\, z], \qquad U_{\text{ale}}(t, c) = \mathbb{E}_{\mathbf{r}}[\mathrm{Var}_z[D_\theta^{\mathbf{r}}(x_t, t, c)]].$$

U_{epi} captures "the model/sub-network is undecided" (reducible, epistemic); U_{ale} captures "the data are genuinely ambiguous" (irreducible, aleatoric). This is the strict decomposition that the seed-only formulation could not claim: with only seeds one cannot disentangle model ambiguity from data stochasticity, whereas two separately controllable noise sources make the split operational.

Conceptual reframing across timesteps (now grounded, not just descriptive).

- **Mode selection** (early steps, high σ): seed variance is high because different z lead to qualitatively different global structures — different poses, backgrounds, object arrangements.

- **Within-mode variability** (late steps, low σ): seed variance is low; the global mode is fixed and only fine texture varies — inherent data stochasticity, not model ambiguity.

With the decomposition above, this reframing can be tied directly to U_{epi} vs U_{ale} rather than asserted qualitatively.

Related Work

- **TREAD**(Krause et al., 2025) — token routing for efficient DiT training; randomly routes tokens around contiguous layer blocks during training, yielding large convergence speedups and improved FID. Presented purely as an efficiency method; the induced stochastic-depth training (and its use for UQ) is not analyzed. We repurpose its routing as an inference-time epistemic mechanism.
- **Generative Uncertainty in Diffusion Models** (Jazbec et al., 2025, NeurIPS/UvA) — proposes Bayesian UQ via last-layer Laplace approximation for U-Net and flow matching models. Does not study DiT architecture, does not analyze per-timestep uncertainty profiles, and explicitly marks aleatoric uncertainty as future work.
- **HyperDM** (Chan et al., NeurIPS 2024) — estimates both epistemic and aleatoric uncertainty via a hyper-diffusion ensemble approach. Methodologically rigorous but computationally heavy. Does not provide timestep-resolved analysis.
- **Timestep-Aware Block Masking** (2025) — shows that different DiT blocks activate differently across timesteps, supporting the hypothesis that early and late steps are functionally distinct.
- **Attention entropy in transformers** — well-established as an uncertainty proxy in NLP and vision transformers; not yet applied to diffusion timestep analysis.

The proposed combination — routing variance (TREAD-induced stochastic depth) as a legitimate epistemic proxy, combined with seed variance and attention entropy, analyzed per-timestep in DiT, decomposed into epistemic/aleatoric components, and linked to generation quality — is not covered by any of the above works.

Specific Tasks

Task 1 — Implement the Three Uncertainty Proxies

Using a pretrained TREAD DiT-XL/2 and a vanilla DiT-XL/2:

1. Implement ensemble variance $U_{\text{ens}}(t, c)$: for each class label c and each timestep bin, run $K = 16$ full-path forward passes with different noise seeds; compute per-patch variance of $\hat{x}_0$ predictions.
2. Implement routing variance $U_{\text{route}}(t, c)$ on the TREAD model: at fixed seed z, draw $M = 16$ random routes (matching the training selection rate and the recommended route span $i \approx 2$, $j \approx B - 4$); compute per-patch variance of $\hat{x}_0$ predictions across routes.
3. Implement attention entropy $U_{\text{attn}}(t, \ell)$: hook into each DiT attention block, extract attention maps, compute Shannon entropy per head, average across heads and patches.
4. Log all three metrics across all timestep bins for a set of 50 ImageNet classes (covering easy, medium, hard classes by expected FID contribution).

Task 2 — Validate Routing Variance as an Epistemic Signal

This task is the linchpin: the epistemic/aleatoric decomposition is only justified if U_{route} behaves like epistemic uncertainty and is not redundant with U_{ens}.

1. **Reducibility / OOD sensitivity:** compute U_{route} for in-distribution ImageNet classes vs OOD / hard classes; an epistemic signal should be *higher* where the model is less certain (OOD, rare, hard-FID classes).
2. **Capacity / data trend:** compare U_{route} across DiT-B/L/XL TREAD checkpoints (and across training-iteration checkpoints if available); epistemic uncertainty should *decrease* with capacity and training.
3. **Non-redundancy:** measure correlation between U_{route} and U_{ens} per timestep; low correlation supports that they carry different information (sub-network vs data noise).
4. **Inference-time route sanity:** verify that activating routes at inference (rather than the standard full-path inference) does not catastrophically degrade $\hat{x}_0$, so that U_{route} is meaningful; report the route configurations for which this holds.

Deliverable: Evidence table (OOD vs ID, capacity trend, U_{route}–U_{ens} correlation) + route-sanity ablation.

Task 3 — Timestep Profiles and Epistemic/Aleatoric Decomposition

Plot and analyze:

- $U_{\mathrm{ens}}(t)$, $U_{\mathrm{route}}(t)$, and $U_{\mathrm{attn}}(t, \ell)$ as functions of t — do they peak early and decay? Do decay shapes differ by class and by proxy?
- Decomposition: estimate $U_{\mathrm{epi}}(t, c)$ and $U_{\mathrm{ale}}(t, c)$ via the law of total variance (nested sampling over routes $\mathbf{r}$ and seeds z); plot how the epistemic vs aleatoric share evolves across timesteps.
- Mode transition point: find the timestep t^* where $U_{\mathrm{ens}}(t)$ (and, separately, $U_{\mathrm{epi}}(t)$) drops most steeply — a data-driven estimate of when the model commits to a global structure.

Key comparison: simple classes (e.g., "lemon") vs ambiguous ones (e.g., "crane" — bird vs machine) — the ambiguous class should show a delayed or more gradual t^* and a larger early-step epistemic share.

Deliverable: Per-class timestep profiles for all proxies, epistemic/aleatoric share curves, and estimated t^* values.

Task 4 — Spatial Uncertainty Heatmaps

Using patch-level variance ($U_{\mathrm{ens}}(t, p)$ and $U_{\mathrm{route}}(t, p)$) for spatial patch p:

1. Generate heatmaps of per-patch uncertainty at different timestep bins (early / mid / late), separately for seed variance and routing variance.
2. Compare heatmaps to semantic segmentation masks (pretrained SAM or DINO-based segmenter) — test whether high-uncertainty patches co-localize with semantically ambiguous regions (object boundaries, backgrounds, occluded areas), and whether epistemic (routing) and aleatoric (seed) maps localize differently.
3. Visualize for 10–15 representative images per class.

Note: qualitative/exploratory task — the goal is compelling visual evidence, not a quantitative benchmark.

Deliverable: Heatmap visualization grid (seed vs routing) + qualitative analysis.

Task 5 — Link Uncertainty to Per-Sample Quality

Use per-sample quality metrics (FID is distributional and unsuitable for per-sample correlation):

- CLIP score — text/class-image alignment
- Inception score contribution — per-sample confidence of Inception-v3 classifier
- Intra-class LPIPS — diversity of generations for the same class

For each class c, compute early/late aggregates (top/bottom 20% σ bins) of each proxy: $U_{\text{ens}}^{\text{early/late}}$, $U_{\text{route}}^{\text{early/late}}$, and the decomposed $U_{\text{epi}}^{\text{early/late}}$, $U_{\text{ale}}^{\text{early/late}}$. Compute Spearman correlation of each with CLIP score and IS contribution across classes.

Hypotheses: (i) early-step uncertainty predicts quality better than late-step uncertainty, because early-step uncertainty determines whether the model commits to a coherent mode; (ii) the *epistemic* early-step component $U_{\text{epi}}^{\text{early}}$ is a stronger and cleaner predictor of low quality than raw seed variance, since it isolates model ambiguity from data stochasticity.

Deliverable: Correlation table (early vs late $\times$ proxy/component $\times$ quality metric) + scatter plots.